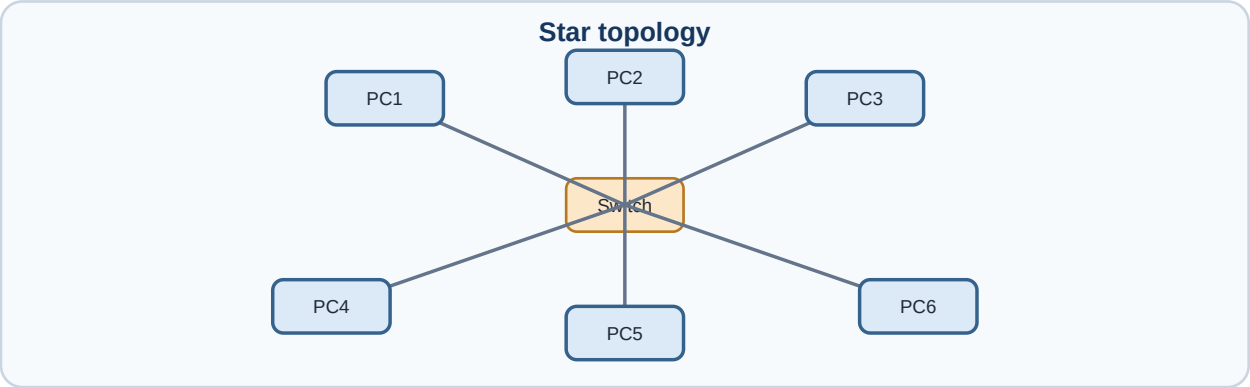
NETWORK TOPOLOGIES

Question 1 — Diagrams and real-world examples

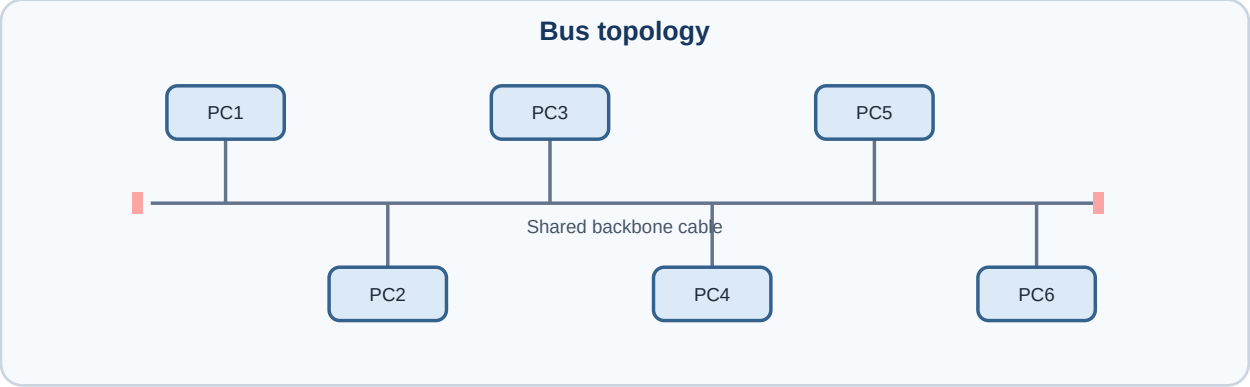
A network topology is the arrangement of devices and connections in a computer network. The diagrams below are simplified examples; real services may use several topologies and cloud data centres together.

Star topology



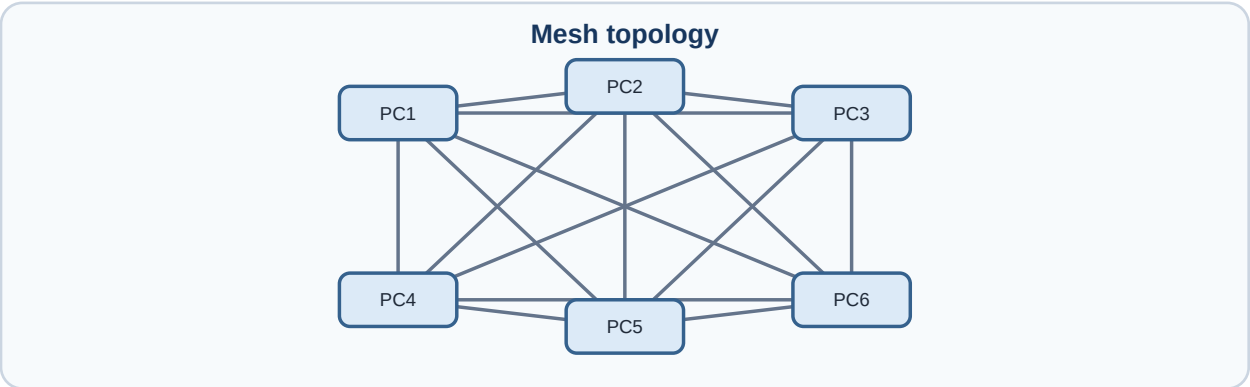
All devices connect to a central switch or hub. Example: a Flipkart office or warehouse LAN. A switch sends order and inventory data to the correct computer, and a single PC cable failure usually affects only that PC.

Bus topology



All devices share one main backbone cable. Example: a small legacy lab network. It uses less cable, but a fault in the backbone can interrupt the whole network; modern Ethernet LANs generally prefer switches.

Mesh topology



Devices have multiple direct links between them. Example: a mesh Wi-Fi system used to carry WhatsApp calls or video traffic around a home. Multiple paths can keep traffic moving if one link fails; internet apps themselves do not run only on a mesh topology.

Question 2 — Comparison of six common topologies

There is no single topology that is best for every situation. The best choice depends on network size, budget, reliability needs, and how easily the network must be expanded.

Topology	Four advantages	Four disadvantages	Best use case / app example
Star	• Easy to install • simple to troubleshoot • one cable/device fault usually affects only one endpoint • easy to add computers.	• Needs a central switch • more cable than bus • switch failure can stop the LAN • switch ports limit expansion.	Office, school, or warehouse LAN. Example: computers updating Zomato-style orders through a central switch.
Bus	• Low cable use • inexpensive for a tiny setup • no central switch needed • simple layout.	• Backbone failure can stop all devices • performance drops with heavy traffic • fault location can be difficult • limited distance/device capacity.	Legacy or learning-lab demonstration. Example: explaining how several clients share one communication line.
Ring	• Predictable traffic flow • orderly access can reduce collisions • each node has a defined neighbour • can work well in controlled industrial links.	• A break can disrupt a basic ring • adding/removing nodes can interrupt service • troubleshooting may take time • less common in modern office LANs.	Specialised or legacy ring networks. Example: a campus/industrial network where predictable traffic is important.
Mesh	• Multiple paths improve resilience • can reroute around a failed link • no single link must carry all traffic • useful for wireless coverage.	• Expensive to build fully • needs many links/ports • configuration is more complex • managing and securing many paths takes effort.	Wireless mesh Wi-Fi or critical links. Example: keeping a video call connected as traffic takes another available path.
Tree	• Scales by branches • supports departments/sections • easier to organise than one huge flat LAN • can centralise management.	• Root/core failure can affect many branches • more switches and cabling cost more • configuration is more complex • bad design can create bottlenecks.	Large school, office, or multi-floor building. Example: separate branches for billing, inventory, and support systems.
Hybrid	• Combines strengths of different layouts • flexible design • can scale by department • reliability can be improved with redundant links.	• Planning is more complex • higher equipment cost • troubleshooting may require multiple skills • compatibility/configuration must be managed.	Medium/large business network. Example: a company using star LANs in each office and redundant links between buildings.

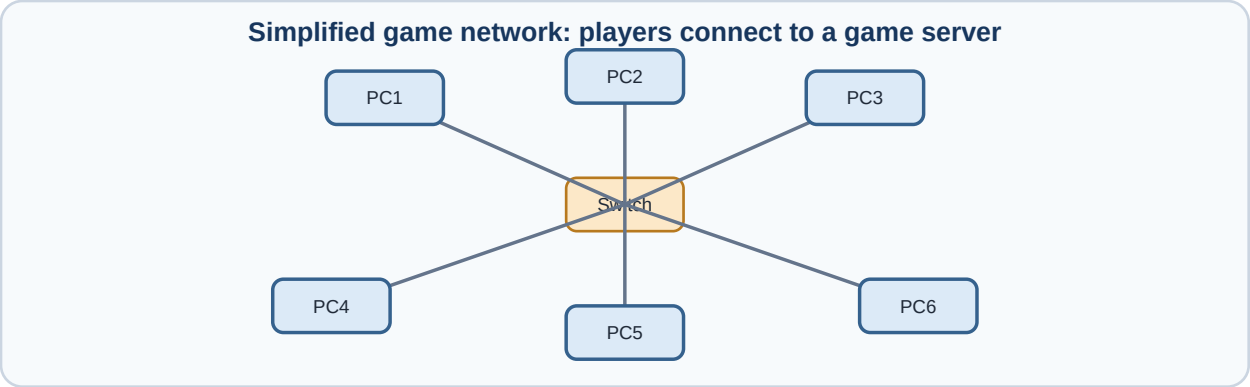
Overall best for most small and medium wired LANs: Star topology. It is easy to set up, expand, and troubleshoot, and one computer cable failing does not normally disconnect the other computers. The central switch is the main weakness, so larger or critical networks may add redundant switches/links or use a hybrid design.

Question 3 — Topology for a multiplayer mobile game

Recommended design: a client–server system arranged like a star around game servers, with a hybrid network behind the servers.

Players' phones should send game actions to a regional game server rather than connecting directly to every other player. The server validates movement, hits, and match state, then sends updates back to players. This reduces unnecessary traffic and makes cheating harder than a peer-to-peer design.

Speed: placing servers close to players and using suitable regional matchmaking helps reduce latency, which is important in games like PUBG or Free Fire. Reliability: multiple servers and backup network paths can keep matches available if one server or link fails. Cost: this requires server and hosting expenses, but it is easier to manage and scale than a full mesh in which every player connects to every other player.



The diagram is conceptual: real games may use load balancers, multiple regional servers, databases, and redundant links. A pure full-mesh topology is not suitable for a large mobile battle game because the number of connections grows quickly as players are added.

Step-by-step design

• Players connect to the nearest available regional game server.

• The server receives and checks player actions, then updates the match state.

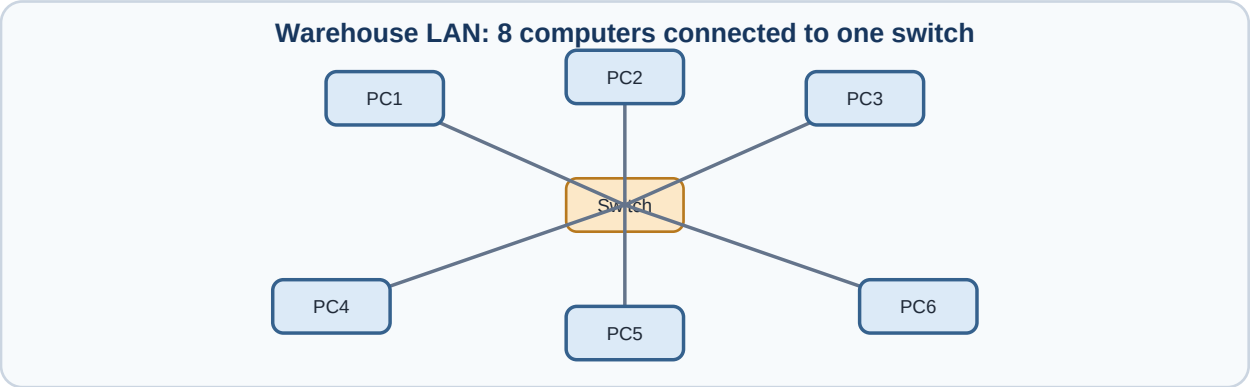
• The server distributes relevant updates to the players in that match.

• A matchmaking/load-balancing layer sends new matches to available servers.

• Monitoring, backups, and redundant links help recover from failures.

Question 4 — Network for a small warehouse with 8 computers

Recommendation: Star topology using one central Ethernet switch. Each of the eight computers connects to a separate switch port using an Ethernet cable. The switch can also connect to the router/internet and, if needed, a shared printer or local server.



Why this is suitable

A star network is affordable for a small warehouse and is simple to install. If one computer's cable is damaged, the other computers can continue working. Troubleshooting is straightforward because staff can check that computer's cable and switch port. Choose a switch with at least 8 usable ports for the computers, plus extra ports for the router and future devices; a 16-port switch provides useful room to expand.

Step-by-step setup

• Place the Ethernet switch in a safe, ventilated location near the network equipment.

• Connect each of the 8 computers to the switch with a separate Ethernet cable.

• Connect the switch to the router so the computers can reach the internet or cloud inventory system.

• Give each computer a valid network configuration (usually automatically by DHCP).

• Test access to the order-processing and inventory application from every computer.

• Label the cables and record the switch port used by each computer to make future troubleshooting easier.

Conclusion: For this warehouse, star topology is the best practical choice because it balances cost, reliability, easy troubleshooting, and future expansion. If the warehouse grows, it can add more switches or use a hybrid/tree layout.